

Challenge (Habanero)

- Q1.** A hiker is making trail mix with $6x + 12$ ounces of nuts and $9x + 18$ ounces of dried fruit. What is the greatest common factor of these two expressions?
(A) $3(x + 2)$
(B) $3x + 6$
(C) $6x + 12$
(D) $3(2x + 6)$
(E) $6(x + 2)$
- Q2.** A wildlife reserve has $15x - 30$ animals in one area and $25x - 50$ animals in another area. What is the greatest common factor of these two expressions?
(A) $5(x - 2)$
(B) $5x - 10$
(C) $5(3x - 6)$
(D) $10x - 20$
(E) $15(x - 2)$
- Q3.** A car is traveling $24x + 48$ miles per hour and using $16x + 32$ gallons of fuel per hour. What is the greatest common factor of these two expressions?
(A) $8(x + 2)$
(B) $8x + 16$
(C) $8(3x + 6)$
(D) $16x + 32$
(E) $16(x + 2)$
- Q4.** A group of friends want to hike $18x - 36$ feet uphill and $22x - 44$ feet downhill. What is the greatest common factor of these two expressions?
(A) $2(x - 2)$
(B) $2x - 4$
(C) $2(9x - 18)$
(D) $9x - 18$
(E) $2(11x - 22)$
- Q5.** A park ranger is monitoring $12x + 24$ bird species in one area and $15x + 30$ bird species in another area. What is the greatest common factor of these two expressions?
(A) $3(x + 2)$
(B) $3x + 6$
(C) $3(4x + 8)$
(D) $6x + 12$
(E) $3(5x + 10)$
- Q6.** A hiking group is consuming $8x + 16$ liters of water per hour and $12x + 24$ calories per hour. What is the greatest common factor of these two expressions?
(A) $4(x + 2)$
(B) $4x + 8$
(C) $4(2x + 4)$
(D) $8x + 16$
(E) $4(3x + 6)$
- Q7.** A nature reserve has $20x - 40$ acres of forest and $25x - 50$ acres of grassland. What is the greatest common factor of these two expressions?
(A) $5(x - 2)$
(B) $5x - 10$
(C) $5(4x - 8)$
(D) $10x - 20$
(E) $5(5x - 10)$
- Q8.** A group of hikers want to travel $30x + 60$ miles and use $20x + 40$ gallons of fuel. What is the greatest common factor of these two expressions?
(A) $10(x + 2)$
(B) $10x + 20$
(C) $10(3x + 6)$
(D) $20x + 40$
(E) $10(2x + 4)$

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $12x^2 + 24x + 36$ by factoring out the greatest common factor. Show all steps and explain your reasoning.
- Q10.** Factor out the greatest common factor from the expression $15x^3 - 30x^2 + 45x$. Then, simplify the resulting expression and explain the process.

Chef's Tips

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- Tip 1: Emphasize the importance of identifying the greatest common factor in each expression.
- Tip 2: Encourage students to use the distributive property to simplify expressions after factoring.
- Tip 3: Chef's Tips